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Editorial Board
IEEE Transactions on Parallel and Distributed Systems (TPDS)
IEEE Computer Society

Dear Editorial Board,

I am writing to formally submit my manuscript entitled **"Subtractive Mesh Topologies: Federated Decentralized Processing in Edge AI"** for consideration as a full-length research article in *IEEE Transactions on Parallel and Distributed Systems*.

Current deployment paradigms for generative Artificial Intelligence heavily rely on geometrically centralized architectures, inherently suffering from asynchronous dependency limits and privacy vulnerabilities mapped in classic Byzantine scenarios. This paper constructs a mathematical alternative: a completely decentralized, zero-knowledge topology designed specifically for edge-native hardware matrices.

By bounding inference gradients to single endpoints and broadcasting exclusively deterministic cryptographic hashes, our derived algorithmic models empirically prove that system-wide intelligence aggregation can occur sequentially in absolute $O(1)$ constant time, slashing bandwidth constraints by 96%. The paper strictly bounds these mechanisms to thermodynamic parsimony controls and accepted distributed synchronization axioms.

Because this paper offers a rigorous algorithmic solution strictly targeted at distributed edge networking environments and large-scale decentralized systems, I believe it aligns flawlessly with TPDS's focus on decentralized algorithm architectures and parallel grid optimizations.

This manuscript is original, formally verified, and not under consideration elsewhere.

I appreciate the Editorial Board's rigorous evaluation of our mathematical formalizations.

Sincerely,

Rafael D. De Paz